SE444 Mid-Term II – Answer Key & Grading Guide (Fall 2025)

Part I: Multiple Choice (4 pts)

Option	Q1 (1pt)	Q2 (1pt)	Q3 (1pt)	Q4 (1pt)
A	**X**		**X**	
B			**X**	**X**
C	**X**	**X**	**X**	
D				**X**
E			**X**	

Answers: Q1: A,C — Q2: C — Q3: A,B,C,E — Q4: B,D **Grading:** 1pt if all correct; 0.5pt if 1 error; 0pt if 2+ errors.

Part II: Logical Reasoning (10 pts)

Q II.1: FOL Translation (3 pts)

Answer:

$$\forall s, t, c\,(\text{Student}(s) \land \text{Teacher}(t) \land \text{Course}(c)$$
$$\land \text{Takes}(s, c) \land \text{Teaches}(t, c)) \rightarrow$$
$$(\exists u\,(\text{Topic}(u) \land \text{Covers}(c, u) \land \text{Understands}(s, u)))$$

Rubric: 3pts correct; 2pts minor errors; 1pt partial; 0pts wrong structure.
Key: $\forall s, t, c$; implication; $\exists u$; all predicates.

Q II.2: FOL with Negation (3 pts)

Answer (accept any equivalent):
A: $\forall t, s\,((\text{Teacher}(t) \land \text{Student}(s) \land \text{GradedBy}(s, t)) \rightarrow (\exists c\,(\text{Course}(c) \land \text{Teaches}(t, c) \land \text{Takes}(s, c))))$
B: $\forall t, s\,(\text{Teacher}(t) \land \text{Student}(s) \land \neg\exists c\,(\text{Course}(c) \land \text{Teaches}(t, c) \land \text{Takes}(s, c)) \rightarrow \neg\text{GradedBy}(s, t))$
C: $\neg\exists t, s\,(\text{Teacher}(t) \land \text{Student}(s) \land \text{GradedBy}(s, t) \land \forall c\,((\text{Course}(c) \land \text{Teaches}(t, c)) \rightarrow \neg\text{Takes}(s, c)))$
Rubric: 3pts any equivalent form; 2pts minor errors; 1pt correct idea; 0pts wrong.

Q II.3: CNF Conversion (4 pts)

Steps (1pt each):
1. Negate: $\neg\forall x\,(\text{Student}(x) \rightarrow \text{Hardworking}(x))$
2. Move $\neg$ in: $\exists x\,\neg(\text{Student}(x) \rightarrow \text{Hardworking}(x))$
3. Eliminate $\rightarrow$ (De Morgan): $\exists x\,(\text{Student}(x) \land \neg\text{Hardworking}(x))$
4. Skolemize: $\text{Student}(a) \land \neg\text{Hardworking}(a)$
Rubric: 1pt/step (must show transformation + rule); 0.5pt if correct but rule not stated.

Part III: CSP (6 pts)

Q III.1: Formulation (2 pts)

Variables (0.5pt): L_1, L_2, L_3
Domains (1pt):
- $D(L_1) = \{A, C\}$
- $D(L_2) = \{A, B, C\}$
- $D(L_3) = \{B, C\}$

Constraints (0.5pt):
- $L_1 \neq L_2$ (overlap 8:30–9:00)
- $L_2 \neq L_3$ (overlap 9:00–9:30)

Q III.2: AC-3 Algorithm (3 pts)

Answer: NO PRUNING - All domains unchanged
AC-3 Algorithm Execution:
Initial State:
- $D(L_1) = \{A, C\}$
- $D(L_2) = \{A, B, C\}$
- $D(L_3) = \{B, C\}$
- Queue: (L_1, L_2), (L_2, L_1), (L_2, L_3), (L_3, L_2)

Step-by-step Arc Processing:
Arc 1: $(L_1 \to L_2)$ **- Constraint** $L_1 \neq L_2$
- Check $L_1 = A$: Needs $L_2 \neq A \to L_2 = B, C$ ✓Supported
- Check $L_1 = C$: Needs $L_2 \neq C \to L_2 = A, B$ ✓Supported

Result: No values pruned from $D(L_1)$
Arc 2: $(L_2 \to L_1)$ **- Constraint** $L_1 \neq L_2$
- Check $L_2 = A$: Needs $L_1 \neq A \to L_1 = C$ ✓Supported
- Check $L_2 = B$: Needs $L_1 \neq B \to L_1 = A, C$ ✓Supported
- Check $L_2 = C$: Needs $L_1 \neq C \to L_1 = A$ ✓Supported

Result: No values pruned from $D(L_2)$
Arc 3: $(L_2 \to L_3)$ **- Constraint** $L_2 \neq L_3$
- Check $L_2 = A$: Needs $L_3 \neq A \to L_3 = B, C$ ✓Supported
- Check $L_2 = B$: Needs $L_3 \neq B \to L_3 = C$ ✓Supported
- Check $L_2 = C$: Needs $L_3 \neq C \to L_3 = B$ ✓Supported

Result: No values pruned from $D(L_2)$
Arc 4: $(L_3 \to L_2)$ **- Constraint** $L_2 \neq L_3$
- Check $L_3 = B$: Needs $L_2 \neq B \to L_2 = A, C$ ✓Supported
- Check $L_3 = C$: Needs $L_2 \neq C \to L_2 = A, B$ ✓Supported

Result: No values pruned from $D(L_3)$
Final State: All domains unchanged - AC-3 terminates
Grading Rubric (3 pts):
- **2.0 pts:** Correct answer (no pruning) + shows systematic arc checking (at least 3 arcs)
- **1.5 pts:** Correct answer + shows checking of 2 arcs with support analysis
- **1.0 pt:** Correct answer + minimal arc checking (1 arc) or just states final domains
- **0.5 pt:** Incorrect pruning but shows logical AC-3 process
- **0 pts:** Only final answer without any work shown

Key: Must show arc-by-arc checking process, not just final result.

Q III.3: Valid Solution (1 pt)

Accept ANY valid assignment that satisfies:
- $L_1 \in \{A, C\}$, $L_2 \in \{A, B, C\}$, $L_3 \in \{B, C\}$
- $L_1 \neq L_2$ (Labs 1&2 overlap)
- $L_2 \neq L_3$ (Labs 2&3 overlap)

All Valid Solutions:

Solution	L_1	L_2	L_3
1	A	B	C
2	A	C	B
3	C	A	B
4	C	B	C

Grading:
- **1 pt:** Any valid assignment
- **0 pts:** Invalid (violates constraints or domains)

Verification: Check domains + $L_1 \neq L_2$ + $L_2 \neq L_3$